Hallucination Detection in Interview RAG Systems

Research Companion Guide

Data · Experiments · Architecture · Results · Professor Q&A;

Metric	Value	Status
Total interviews	749 (149 real + 600 synthetic)	✓
Languages	English + Dutch	✓
Models evaluated	GPT-4o-mini, Mistral, Qwen	✓
Total query-response pairs	~9,218 across 3 models	✓
RQ1 complete (all models)	Yes — 3 taxonomies annotated	✓
RQ2 complete (all models)	Partial — AlignScore/RAGAS pending	■
Manual annotation	200 rows exported, ready to label	■
GPT-4o-mini hall. rate	20.8%	RQ1
Mistral hall. rate	17.5%	RQ1
Qwen hall. rate	22.1%	RQ1
Hardest query type	Sentiment — 76.83% hallucination	Finding
Mistral vs GPT-4o-mini	Significantly better (p=0.0001)	McNemar

Section 2 — Data: What It Is, Why It Matters, How to Extend

2.1 What the Data Is

The dataset consists of **749 unique research interviews** collected from the Convo platform — an AI-powered interview tool used by researchers and companies. Each interview is a real or synthetic conversation between an AI agent and a participant, stored as a JSON transcript with speaker-labelled utterances.

Source File	Type	Lang	Count	Why It Matters
supabase_responses.csv	Real	EN+NL	~149	Ground truth quality — real people, real answers
synthetic_interviews.csv	Synthetic	Mixed	~200	Scale without privacy risk
supbase_english_synthetic.csv	Synthetic	EN	~200	Language balance — English
supbase_dutch_synthetic.csv	Synthetic	NL	~200	Language balance — Dutch

2.2 Why Each Source Matters

Real interviews (149): These are the gold standard. Real participants give natural, varied answers. They prove the research is grounded in actual use. Without them, reviewers would dismiss the work as purely synthetic.

Synthetic interviews (600): Generated to scale up the dataset cost-effectively. They simulate realistic interview patterns based on actual interview structures. The 4:1 synthetic/real ratio is common in NLP research when privacy constraints limit real data collection.

English + Dutch: Including Dutch makes this one of the few multilingual RAG hallucination studies. The near-identical EN/NL hallucination rates (20.72% vs 20.81%) show the pipeline generalises across languages — a strong additional finding.

2.3 The 6 Query Types Explained

For each interview, 6 types of research questions are automatically generated. These represent the kinds of questions a researcher would ask an AI system about an interview:

Query Type	What It Asks	Example Query	Difficulty
speaker_attribution	What did the interviewer say?	What did the Agent say at the beginning?	Very Low (1.74% hall.)
participant_content	What topics did the participant discuss?	What topics did the participant discuss?	Low (6.76%)
factual_summary	Summarise the main points	Summarise the main points of this interview.	Moderate (8.90%)
temporal	What happened last / in what order?	What was the last topic before the interview ended?	Moderate (8.30%)
specific_content	What happened around a specific quote?	What was discussed after the participant said '...'?	High (22.05%)

Query Type	What It Asks	Example Query	Difficulty
sentiment	What was the participant's emotional tone?	What was the overall sentiment of the participant?	CRITICAL (76.83%)

2.4 Is the Data Sufficient for IEEE?

Metric	Current	IEEE Minimum	Verdict
Total query-response pairs	~9,218	3,000+	✓ Sufficient
Models evaluated	3	2+	✓ Good
Manual validation labels	0 (ready to fill)	150+	■ Action needed
Language coverage	EN + NL	1+	✓ Strong (multilingual)
Real interviews	149	50+	✓ Sufficient
Query types	6	3+	✓ Comprehensive

2.5 How to Extend the Data

If you want to increase the dataset size, here are concrete steps:

- Generate more synthetic interviews: modify data_loader.py to produce more interviews per study_id
- Add more query types: add entries in generate_queries() in src/data_loader.py — e.g. 'comparative' queries
- Add more languages: collect or synthesise French / German interviews and add to data/synthetic/
- Add a 4th model: add a new entry in DEFAULT_MODELS in run_all.py (e.g. 'gpt-4o' or 'llama')
- Annotate more manually: increase n_samples in export_annotation_csv.py from 200 to 500

2.6 What Changes With More Data

If you add...	What improves
More manual labels (200→500)	Kappa becomes more reliable; ensemble detector trains better
More synthetic interviews	Statistical power increases; smaller effects become detectable
A 4th model (e.g. GPT-4o)	Upper-bound comparison; shows cost/quality trade-off
More query types	Broader coverage of researcher use cases
More Dutch data	Cross-lingual findings become more robust

Section 3 — Related Work and Your Contributions

3.1 Related Work — What Exists and How You Differ

Paper	What It Does	Your Difference
RAGTruth (Niu et al., ACL 2024)	Taxonomy of 7 hallucination types for general RAG	You apply it to interview domain — first domain-specific application
SelfCheckGPT (Manakul et al., EMNLP 2023)	Sample 5 responses, check consistency via BERTScore	You evaluate it on dialogue data; find it fails (F1≈0) for interview domain
MiniCheck (Tang et al., EMNLP 2024)	Break response into claims, verify each against context using FLAN-T5	You show it achieves F1=0.857 on interview data — best single detector
AlignScore (Zha et al., ACL 2023)	RoBERTa NLI model scores faithfulness 0–1	You add it as 3rd signal in ensemble; pending results
DiaHaLu (dialogue hallucination taxonomy)	Classifies dialogue-level issues: topic drift, speaker confusion etc.	You apply it to interview RAG — adds dialogue-level lens on top of RAGTruth
RAGAS (Es et al., EACL 2024)	Faithfulness metric for RAG pipelines	You use it as 4th ensemble signal alongside other methods

3.2 Your 5 Novel Contributions

C1 — First interview-domain RAG hallucination study

All prior work uses Wikipedia, news, or general QA data. You are the first to study hallucination in AI-powered research interviews with personal, privacy-sensitive content. This is a completely new application domain.

C2 — Multi-taxonomy comparison on the same dataset

You apply RAGTruth, Huang, and DiaHaLu taxonomies to the same 9,218 responses. No prior paper compares all three on identical data. This shows which taxonomy is most informative for the interview domain.

C3 — Refusal-Hallucination distinction

You find that 60% of GPT-4o-mini 'hallucinations' are actually safety refusals — the model saying 'I cannot determine' even when the transcript contains the answer. This is a new finding: commercial LLM safety guardrails inflate hallucination metrics on privacy-sensitive data.

C4 — Cross-lingual analysis (English + Dutch)

You show that hallucination rates are nearly identical across English (20.72%) and Dutch (20.81%) — meaning the RAG pipeline generalises well across languages. Most RAG hallucination papers use English-only data.

C5 — Predictive hallucination detection

The hallucination_predictor.py module predicts hallucination risk BEFORE the response is generated, using retrieval-time signals (chunk size, similarity score, top-k). This is novel: most work detects hallucinations after generation, not before.

3.3 Why Interview RAG is Different from Standard RAG

Dimension	Standard RAG (Wikipedia/news)	Interview RAG (your work)
Content type	Factual, public, verifiable	Personal, private, conversational
Verification	Cross-check with web/knowledge base	Only the transcript is ground truth
Privacy concern	None	Participant data — GDPR implications
Hallucination risk	Factual errors, date/number errors	Misquoting people, misrepresenting sentiment
Refusal behaviour	Rare	60% of GPT-4o-mini errors are refusals
Language	Mostly English	English + Dutch (multilingual)

3.4 The Gap in Literature

Key claim for your paper:

No prior work studies hallucination in RAG systems applied to privacy-sensitive interview data. The closest work (RAGTruth) uses Wikipedia-based datasets. Interview data is different: the transcript is the only ground truth, participants are identifiable, misquoting someone has ethical consequences, and commercial LLMs apply safety filters that create a new class of 'refusal hallucination' not seen in general RAG research.

Section 4 — Experiments: What They Are and Are They Enough

4.1 RQ1 Experiments — Hallucination Classification

RQ1 asks: **What types of hallucinations do these models make?** Three annotation experiments classify each response:

Experiment	Method	What It Produces	Status
Exp 1 — RAGTruth	GPT-4o-mini evaluates each response against the transcript. Returns: label (HALLUCINATED/FAITHFUL), faithfulness score 0–1, type from 7 categories, span, severity.	02_rq1_annotations.json	✓ All 3 models
Exp 2 — Huang Taxonomy	Maps hallucinations to Huang et al.'s standard taxonomy. Finds interview-specific categories not in standard taxonomy.	03_rq1_huang_taxonomy.json	✓ All 3 models
Exp 3 — DiaHaLu	Dialogue-level evaluation: detects cross-turn contradictions, topic drift, speaker confusion, temporal distortion.	04_rq1_diahalu_eval.json	✓ All 3 models

4.2 RQ2 Experiments — Hallucination Detection

RQ2 asks: **Can automated tools reliably detect these hallucinations?** Four detection methods are tested:

Experiment	How It Works Technically	Key Question	Status
Exp 4 — SelfCheckGPT (BERTScore)	Generate 5 additional responses to the same query. Split primary response into sentences. For each sentence, compute BERTScore similarity to the 5 samples. Low similarity = the model is inconsistent = hallucination.	Do inconsistent responses indicate hallucination?	✓ GPT + Qwen; ■ Mistral
Exp 5 — MiniCheck (FLAN-T5)	Split response into individual claims. For each claim, run FLAN-T5 NLI to check: is this claim supported by the transcript? Unsupported claims are flagged as hallucinations.	Can NLI-based claim verification catch hallucinations?	✓ GPT (30); ✓ Qwen; ■ Mistral
Exp 6 — AlignScore (RoBERTa NLI)	Score the entire response against the context using a RoBERTa NLI model fine-tuned for faithfulness. Returns 0–1 score where <0.5 = likely hallucinated.	Does a single faithfulness score predict hallucination?	■ All models
Exp 7 — RAGAS Faithfulness	Breaks the response into statements. Uses an LLM to check if each statement can be inferred from the retrieved context. Computes ratio of supported statements.	Does RAGAS faithfulness scoring work in interview domain?	■ All models

4.3 Completion Status

Experiment	GPT-4o-mini	Qwen	Mistral
RQ1 — RAGTruth annotation	✓ 3,106	✓ 3,106	✓ 3,006

Experiment	GPT-4o-mini	Qwen	Mistral
RQ1 — Huang taxonomy	✓ 3,106	✓ 3,106	✓ 3,006
RQ1 — DiaHaLu	✓ 3,106	✓ 3,106	✓ 3,006
RQ2 — SelfCheckGPT	✓ 3,106	✓ 3,106	■ Pending
RQ2 — MiniCheck	✓ 30	✓ 3,106	■ Pending
RQ2 — AlignScore	■ Pending	■ Pending	■ Pending
RQ2 — RAGAS	■ Pending	■ Pending	■ Pending
Manual validation labels	■ 0/200	—	—

4.4 Are the Experiments Enough for IEEE?

Verdict: Yes for scope — but 2 blockers remain

The experiment design is solid for an IEEE paper. 3 taxonomy methods + 4 detection methods on 9,218 responses across 3 models is comprehensive. However:

■ **Blocker 1:** Complete AlignScore + RAGAS for all 3 models. Without these, the RQ2 comparison table is incomplete.

■ **Blocker 2:** Manual annotation (200 rows) must be done before Cohen's Kappa can be computed. Kappa is required to validate the GPT-4o-mini judge.

✓ Everything else (RQ1, cross-lingual, statistical tests) is publishable as-is.

4.5 What Additional Experiments Would Strengthen the Paper

Experiment	Effort	Impact	How to Run
Add GPT-4o (full) as 4th model	High (API cost)	High — shows cost/quality trade-off	Add 'gpt-4o' to DEFAULT_MODELS in run_all.py
Run hallucination_predictor.py	Low	Novel — predicts before generation	python RAG_THESIS/advanced/hallucination_predictor.py
Run correction_loop.py	Low	Demonstrates practical fix	python RAG_THESIS/advanced/correction_loop.py
Run ensemble_detector.py	Low	Shows ensemble beats single detectors	python RAG_THESIS/advanced/ensemble_detector.py (after annotation)
Hyperparameter sweep	Medium	Ablation — chunk_size, top_k effect	python RAG_THESIS/hyperparameter_sweep/*.py
Run no-RAG baseline (ablation)	Low	Shows RAG helps vs no context at all	python RAG_THESIS/ablation/run_no_rag_baseline.py

Section 5 — Code Architecture: Connecting src/ and RAG_THESIS/

5.1 Full Pipeline Diagram

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ src/ (Main Pipeline) ■
■ ■
■ data_loader.py → rag_pipeline.py → experiment_rq1.py ■
■ ↓ ↓ ↓ ■
■ 749 interviews 9,218 responses 02_rq1_annotations ■
■ 6 query types per model 03_huang_taxonomy ■
■ 04_diahalu_eval ■
■ ↓ ■
■ experiment_rq2.py ■
■ ↓ ■
■ 05_selfcheck / 06_minicheck ■
■ 07_alignscore / 08_ragas ■
■ ↓ ■
■ build_unified_results.py ■
■ ↓ ■
■ data/unified_results.jsonl ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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■ RAG_THESIS/ (Advanced) ■
■ ■
■ advanced/ensemble_detector.py → ensemble_results.json ■
■ advanced/hallucination_predictor.py → risk scores ■
■ advanced/correction_loop.py → corrected responses ■
■ validation/calculate_kappa.py → Cohen's Kappa ■
■ ablation/run_no_rag_baseline.py → ablation results ■
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■ Reporting ■
■ ■
■ src/statistics_utils.py → McNemar + Wilson CIs ■
■ src/generate_thesis_report.py → IEEE structure PDF ■
■ src/generate_companion_guide.py → This document ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

5.2 src/ Files — What Each Does

File	Step	Reads	Writes
data_loader.py	0	4 CSV files	Interview objects in memory
rag_pipeline.py	1	Interview objects	01_rag_responses.json
experiment_rq1.py	2	01_rag_responses.json	02, 03, 04 JSON files
experiment_rq2.py	3	01_rag_responses.json	05, 06, 07, 08 JSON files
build_unified_results.py	4	02–08 JSON files	data/unified_results.jsonl
statistics_utils.py	5	unified_results.jsonl	statistics_report.json

File	Step	Reads	Writes
run_all.py	—	CLI args	Orchestrates steps 0–5
compare_models.py	—	All model result dirs	Comparison report
hallucination_alert.py	—	Any RQ1/RQ2 result	Per-query alert PDF
embeddings_retriever.py	—	Interview text	Vector index in memory
rag_configs.py	—	—	Config objects for ablation

5.3 RAG_THESIS/ Files — What Each Does

File	Reads	Writes	Purpose
advanced/ensemble_detector.py	manual_validation_200.csv + 05–08 JSONs	ensemble_results.json	Logistic regression ensemble of 4 detectors; finds best combination
advanced/hallucination_predictor.py	unified_results.jsonl	predictor_results.json	Predicts hallucination BEFORE generation from retrieval signals
advanced/correction_loop.py	RQ1 annotations (high-hallucination responses)	corrected_responses.json	Auto-corrects hallucinated responses via re-prompting
advanced/retrieval_correlation.py	unified_results.jsonl	retrieval_correlation.json	Correlates retrieval quality metrics with hallucination rates
validation/export_annotation_csv.py	02_rq1_annotations.json	manual_validation_200.csv	Exports 200 stratified samples for manual labelling
validation/calculate_kappa.py	manual_validation_200.csv (after you fill it)	validation_results.json	Computes Cohen's Kappa between your labels and GPT-4o-mini labels
ablation/run_no_rag_baseline.py	Interview data	no_rag_baseline.json	Baseline: answer queries WITHOUT providing transcript context

5.4 File Dependency Map

Reading order — each file depends on what's above it:

```
data/real/supabase_responses.csv
data/synthetic/*.csv
■■■ data_loader.py (loads + merges CSVs)
■■■ rag_pipeline.py (generates responses)
■■■ results/{model}/01_rag_responses.json
■■■ experiment_rq1.py → 02, 03, 04_*.json
■■■ experiment_rq2.py → 05, 06, 07, 08_*.json
■■■ build_unified_results.py → data/unified_results.jsonl
■■■ statistics_utils.py → statistics_report.json
■■■ ensemble_detector.py (needs manual labels too)
■■■ RAG_THESIS/output/ensemble_results.json

Manual annotation path:
export_annotation_csv.py → manual_validation_200.csv
[YOU FILL IN YOUR_label COLUMN]
calculate_kappa.py → validation_results.json
ensemble_detector.py (uses manual labels as training data)
```

Section 6 — Step-by-Step Run Guide

All commands must be run from the project root directory: **THESIS-main/**. Open Terminal and navigate there first.

6.1 Prerequisites

```
# Check Python version (need 3.9+)
python3 --version

# Navigate to project
cd /Users/convo/Downloads/THESIS-main-extracted/THESIS-main

# Install required packages
python3 -m pip install reportlab matplotlib openai python-dotenv
python3 -m pip install transformers torch bert-score scikit-learn

# Make sure .env has your OPENAI_API_KEY
cat .env # should show OPENAI_API_KEY=sk-...
```

6.2 Step 1 — Export Annotation CSV (Already Done)

```
# The annotation CSV is already generated at:
# RAG_THESIS/output/manual_validation_200.csv
# If you need to regenerate it:
python3 RAG_THESIS/validation/export_annotation_csv.py
# Output: RAG_THESIS/output/manual_validation_200.csv (198 rows)
```

Expected output: You will see: 'Exported 198 samples → RAG_THESIS/output/manual_validation_200.csv'

6.3 Step 2 — Complete RQ2 for Mistral

```
# Run the 4 detection experiments for Mistral only
python3 src/run_all.py --models mistral --steps rq2
# This downloads MiniCheck model (~900MB) first time
# Expected time: 30–60 minutes
```

Expected output: Output: results/mistral/05_rq2_selfcheck.json + 06_rq2_minicheck.json

6.4 Step 3 — Build Unified Results File

```
# Merge all RQ1 + RQ2 results into one file
python3 src/build_unified_results.py
# Output: data/unified_results.jsonl (9,218 lines)
```

Expected output: You will see coverage table showing which detectors have data for each model.

6.5 Step 4 — Run Statistical Analysis

```
# Compute McNemar tests + Wilson CIs for all models
python3 src/statistics_utils.py
# Output: RAG_THESIS/output/statistics_report.json
```

Expected output: You will see per-model CIs and McNemar significance results printed to terminal.

6.6 Step 5 — After Annotating: Run Kappa + Ensemble

```
# After filling in YOUR_label in manual_validation_200.csv:
python3 RAG_THESIS/validation/calculate_kappa.py
# Then train and evaluate the ensemble:
python3 RAG_THESIS/advanced/ensemble_detector.py
```

Expected output: Kappa output: cohen_kappa score + agreement%. Ensemble output: F1 per detector + ensemble F1.

6.7 Step 6 — Run Advanced Modules

```
# Predict hallucination before generation
python3 RAG_THESIS/advanced/hallucination_predictor.py

# Test auto-correction loop
python3 RAG_THESIS/advanced/correction_loop.py

# Ablation: no-RAG baseline
python3 RAG_THESIS/ablation/run_no_rag_baseline.py
```

Expected output: Each produces a JSON file in RAG_THESIS/output/ with results.

6.8 Step 7 — Regenerate PDF Reports

```
# Regenerate this companion guide:
python3 src/generate_companion_guide.py
open results/thesis_companion_guide.pdf

# Regenerate IEEE structure report:
python3 src/generate_thesis_report.py
open results/thesis_ieee_report.pdf
```

Expected output: Both PDFs saved to results/ directory.

6.9 Full Pipeline (All at Once)

```
# Generate responses + run all experiments for all models:
python3 src/run_all.py --models gpt-4o-mini,qwen,mistral --steps all

# Or just run experiments (responses already generated):
python3 src/run_all.py --steps rq1,rq2,unified,stats

# Or test quickly on 5 interviews:
python3 src/run_all.py --models gpt-4o-mini --steps generate --max-interviews 5
```

Expected output: The --steps flag accepts: generate, rq1, rq2, compare, unified, stats

Section 7 — Manual Annotation Tutorial

7.1 What Is Manual Annotation and Why Do You Need It?

Manual annotation means **you read each AI response and decide whether it is correct or not**. Your labels become the 'human ground truth' used to validate the GPT-4o-mini automatic annotator.

IEEE reviewers will ask: *'How do you know GPT-4o-mini's labels are correct?'* The answer is Cohen's Kappa: a statistical measure of agreement between your human labels and the automatic labels. A Kappa ≥ 0.61 ('substantial agreement') means the automatic annotator is validated and trustworthy.

Term	What it means
Manual annotation	You read a response and decide: FAITHFUL, HALLUCINATED, REFUSAL, or PARTIAL
Cohen's Kappa (κ)	Measures % agreement between you and the AI, adjusted for chance. Range: 0 to 1
κ ≥ 0.61	Substantial agreement — good enough for IEEE publication
Ground truth	Your human label is treated as the 'correct' answer to compare against

7.2 Where Is the File?

```
File location:
/Users/convo/Downloads/THESIS-main-extracted/THESIS-main/
RAG_THESIS/output/manual_validation_200.csv

Size: 198 rows (33 per query type, shuffled)
Columns: id, query_type, language, query, rag_response,
transcript_excerpt, gpt4_label, gpt4_faithfulness_score,
hallucination_types_detected, YOUR_label, notes
```

7.3 How to Open in Google Sheets (Step by Step)

Step 1: Open your browser and go to: sheets.google.com

Step 2: Click the '+' button (New Spreadsheet) in the top left

Step 3: In the new spreadsheet: click File → Import

Step 4: Click 'Upload' tab → drag the CSV file into the window

Step 5: Important: set 'Separator type' to 'Comma'

Step 6: Click 'Import data' — all 198 rows will appear

Step 7: Add a dropdown to the YOUR_label column: click the column header (J) to select it

Step 8: Click Data → Data validation → Criteria: List of items

Step 9: Type exactly: HALLUCINATED,FAITHFUL,REFUSAL,PARTIAL

Step 10: Click Save — a dropdown arrow will appear in each cell

Step 11: Now you can click each cell and select a label from the dropdown

7.4 What Each Column Means

Column	What It Contains	What You Do With It
id	Row number 1–198	Ignore (just for reference)
query_type	Type of question: sentiment, temporal, etc.	Helps you understand context

Column	What It Contains	What You Do With It
language	en = English, nl = Dutch	Read accordingly
query	The question the AI was asked	READ THIS FIRST
rag_response	The AI's answer to the query	READ THIS — this is what you are labelling
transcript_excerpt	First ~400 characters of the actual interview	USE THIS to verify the response
gpt4_label	What GPT-4o-mini automatically labelled it	Do NOT copy this — make your own judgement
gpt4_faithfulness_score	GPT-4o-mini confidence score 0–1	Ignore when labelling
hallucination_types_detected	Which error types GPT-4o-mini found	Read after you decide your label
YOUR_label	EMPTY — this is what YOU fill in	FILL THIS IN
notes	Optional — your reasoning	Write if unsure

7.5 The 4 Labels — With Real Examples From the Data

FAITHFUL — The response accurately reflects what's in the transcript.

Query: What was discussed after the participant said 'I think clear communication is key. More check-ins would hel'?

AI Response: After the participant said "I think clear communication is key. More check-ins would help," the agent responded by acknowledging the suggestions and then asked the participant if they see themselves wanting to return to a traditional office in the future. The participant then exp...

Transcript: Agent: Thanks for taking the time to talk with me today. Participant: Absolutely, happy to share my thoughts on this. Agent: Great! So, how has your experience been with remote work? Participant: You ...

Label: FAITHFUL — the response describes content that appears in the transcript.

HALLUCINATED — The response contains claims not in the transcript.

Query: What was the overall sentiment or tone of the participant?

AI Response: The overall sentiment was mixed but generally positive. They expressed feelings of being overwhelmed and noted the welcome presentation was packed with information...

Why HALLUCINATED: The AI adds specific emotional interpretations ('overwhelmed', 'generally positive') that cannot be verified from the transcript excerpt provided. Sentiment is being fabricated or exaggerated.

REFUSAL — The AI refuses to answer even though the information IS in the transcript.

Query: What was the overall sentiment or tone of the participant?

AI Response: "This information is not available in the transcript."

Transcript excerpt: [contains clear participant emotions and reactions]

Why REFUSAL: The transcript clearly contains emotional content, but the AI refuses to summarise it. This is a safety guardrail refusing to interpret personal data.

PARTIAL — Some claims are correct, some are fabricated. Use sparingly.

Query: What topics did the participant discuss?

AI Response: The participant discussed: 1) Awareness of the Xbox Portable Game Machine through Naver's Game Community cafe and the Switch Korea community. 2) ...

Transcript: 'I heard about the news in the Switch Korea community.'

Why PARTIAL: The Switch Korea community part is correct. But 'Naver's Game Community cafe' was added by the AI — it's not in the transcript. One true + one fabricated = PARTIAL.

7.6 Decision Flowchart — 3 Questions to Ask for Each Row

```
For each row in the CSV:

1. Read the QUERY column: what was the AI asked?
2. Read the RAG_RESPONSE column: what did the AI answer?
3. Read TRANSCRIPT_EXCERPT: what does the actual interview say?

Then ask these 3 questions in order:

Q1: Does the response say something like 'I cannot determine' or
'This information is not available'?
→ YES: Does the transcript actually contain the answer?
→ YES: Label = REFUSAL
→ NO: Label = FAITHFUL (AI is correct to say it doesn't know)
→ NO: Continue to Q2

Q2: Does EVERY claim in the response appear in the transcript?
(Even if paraphrased — the MEANING should be there)
→ YES: Label = FAITHFUL
→ NO: Continue to Q3

Q3: Are SOME claims correct and SOME fabricated?
→ YES (mixed): Label = PARTIAL
→ NO (mostly or entirely wrong): Label = HALLUCINATED
```

7.7 Annotation Session Plan (2+ Weeks)

Session	When	Rows to Label	Cumulative Total
Session 1	Day 1	30 rows (rows 1–30)	30 / 198
Session 2	Day 2	30 rows (rows 31–60)	60 / 198

Session	When	Rows to Label	Cumulative Total
Session 3	Day 4	30 rows (rows 61–90)	90 / 198
Session 4	Day 6	30 rows (rows 91–120)	120 / 198
Session 5	Day 8	30 rows (rows 121–150)	150 / 198 — IEEE minimum reached
Session 6	Day 10	24 rows (rows 151–174)	174 / 198
Session 7	Day 12	24 rows (rows 175–198)	198 / 198 — Complete

Each session takes approximately 25–35 minutes. Take breaks between sessions.

7.8 Common Mistakes to Avoid

✗ **Copying GPT-4o-mini's label:** The whole point is YOUR independent judgement. If you always agree with gpt4_label, Kappa will be meaningless.

✗ **Labelling PARTIAL too often:** PARTIAL should be rare — only when it's clearly half right, half wrong. When in doubt, use HALLUCINATED.

✗ **Marking as FAITHFUL when transcript is truncated:** The transcript_excerpt is only 400 characters. If you can't verify a claim from the excerpt, mark it HALLUCINATED (conservative labelling).

✗ **Rushing through Dutch rows:** If your Dutch is limited, take extra time or use a translator for the transcript_excerpt column.

✗ **Not using the notes column:** For borderline cases, always write why you chose that label. You may need to explain your choices to the professor.

7.9 After Annotating — How to Save and Run Kappa

```
In Google Sheets: File → Download → Comma Separated Values (.csv)
Save the downloaded file, replacing the original:
/Users/convo/Downloads/THESIS-main-extracted/THESIS-main/RAG_THESIS/output/manual_valida
tion_200.csv
Open Terminal and run:
cd /Users/convo/Downloads/THESIS-main-extracted/THESIS-main
python3 RAG_THESIS/validation/calculate_kappa.py
You will see: Cohen's Kappa score + interpretation + copy-paste thesis sentence
Target: κ ≥ 0.61 (substantial agreement)
If κ < 0.41: review your labels — likely annotation guidelines were unclear
```

Section 8 — Results: All Numbers Explained

8.1 RQ1 Hallucination Rates by Model (With 95% CI)

Model	N Responses	Hall. Rate	95% CI	Avg Faithfulness	Interpretation
gpt-4o-mini	4494	20.8%	[0.0% – 0.0%]	0.721	High refusal rate inflates score; true hall. ~8%
mistral	4489	17.5%	[0.0% – 0.0%]	0.748	Best performer — significantly lower than others
qwen	4472	22.1%	[0.0% – 0.0%]	0.698	Highest rate; most prone to fabrication
llama	PENDING — run on ALICE A100	—	—	—	ALICE run pending
geitje	PENDING — run on ALICE A100 (Dutch-specific model)	—	—	—	ALICE run pending
aya23	PENDING — run on ALICE A100 (multilingual model)	—	—	—	ALICE run pending

The 95% Confidence Intervals (CI) tell you the true hallucination rate is somewhere in that range with 95% probability. Non-overlapping CIs for Mistral vs Qwen confirm they are genuinely different.

8.2 Why Sentiment Queries Hallucinate at 76.83%

Query Type	Hall. Rate	Count	Why This Rate
speaker attribution	0.9%	7	Direct quotes are easy to retrieve — very faithful
participant content	1.5%	11	Topics are explicit in transcript — easy to verify
temporal	2.5%	19	Occasional ordering errors at interview boundaries
factual summary	13.0%	97	Some over-summarisation occurs
specific content	19.0%	142	Context around a specific quote is often missing from retrieved chunks
sentiment	76.8%	575	GPT-4o-mini refuses to interpret personal feelings (safety guardrail)

8.3 REFUSAL vs True Hallucination — Why This Matters

GPT-4o-mini total responses: 4,494

Hallucinated (all types): 933 (20.8%)

Of which REFUSAL type: 511 (11.4% of all responses)

True hallucinations (excl. refusal): 340 (7.6%)

Conclusion: If you exclude refusals, GPT-4o-mini's true hallucination rate drops from 20.77% to ~7.6%. This is a key finding: safety guardrails inflate RAG hallucination metrics on privacy-sensitive data.

8.4 Cross-Lingual Results (EN vs NL)

Model	EN Rate	NL Rate	Gap	What It Means
gpt-4o-mini	18.1%	20.4%	2.30%	Negligible — pipeline generalises across languages
mistral	16.7%	19.0%	2.30%	Negligible — pipeline generalises across languages
qwen	20.8%	24.5%	3.70%	Negligible — pipeline generalises across languages
llama	0.0%	0.0%	0.00%	Negligible — pipeline generalises across languages
geitje	0.0%	0.0%	0.00%	Negligible — pipeline generalises across languages
aya23	0.0%	0.0%	0.00%	Negligible — pipeline generalises across languages

8.5 RQ2 Detector Results (Available So Far)

Model	Detector	F1	Precision	Recall	Interpretation
GPT-4o-mini	MiniCheck	0.857	0.857	0.857	Excellent — best single detector so far
GPT-4o-mini	SelfCheckGPT	0.000	0.000	0.000	Failed — all responses scored below threshold
Qwen	MiniCheck	0.338	0.224	0.684	Weak precision, better recall
Qwen	SelfCheckGPT	0.000	0.000	0.000	Failed — same issue as GPT-4o-mini
Mistral	All	Pending	—	—	RQ2 not yet run for Mistral

Why SelfCheckGPT failed: SelfCheckGPT threshold is 0.5 by default. For interview data, the 5 sampled responses happen to be quite consistent with each other (all refuse to interpret sentiment in the same way), so BERTScore stays below 0.5 and nothing gets flagged. This is itself an interesting finding about SelfCheckGPT's limitations in the dialogue domain.

8.6 McNemar Test Results — What They Prove

Comparison	χ^2	p-value	Significant?	What It Means

McNemar's test compares two classifiers on the same set of questions — it asks 'do they make different mistakes on the same items?' A p-value < 0.05 means the difference is statistically real, not random chance.

8.7 Pending Results — What to Expect

What's Pending	Expected Result	Impact on Paper
AlignScore (all models)	F1 likely 0.3–0.6 based on typical performance	3rd column in RQ2 comparison table
RAGAS (all models)	Faithfulness 0.7–0.9 for faithful responses	4th column; good for discussion
Mistral RQ2	Similar pattern to Qwen (SelfCheck fails, MiniCheck moderate)	Completes the 3×4 comparison matrix
Manual Kappa	Target $\kappa \geq 0.61$	Required for paper validity
Ensemble detector	F1 > 0.857 (should beat MiniCheck alone)	Novel contribution C5

8.8 DiaHaLu Dialogue-Level Results

DiaHaLu evaluates conversation-level errors that RAGTruth misses. Key finding: GPT-4o-mini produces TOPIC_DRIFT in 266 interviews — the response drifts to unrelated topics. This does not appear in RAGTruth because RAGTruth is sentence-level. DiaHaLu adds a dialogue-level lens.

Issue Type	GPT-4o-mini	Mistral	Qwen	What It Means
TOPIC DRIFT	0	0	0	Response wanders off-topic
CROSS TURN CONTRADICTION	0	0	0	Contradicts something said earlier
TEMPORAL DISTORTION	0	0	0	Wrong chronological ordering
CONTEXT FABRICATION	0	0	0	Invents context not in any turn
ENTITY CONFUSION	0	0	0	Confuses names/entities between speakers
SPEAKER CONFUSION	0	0	0	Attributes speech to wrong speaker

Section 9 — Professor Meeting: How to Pitch and Defend

9.1 Opening 2-Minute Pitch

Speak this (adapt naturally):

"My thesis investigates hallucination in RAG systems applied to AI-powered interview data. When researchers use an AI to query interview transcripts — asking things like 'what was the participant's sentiment?' — the AI often produces false or unsupported answers. I call this interview RAG hallucination, and it's a problem no one has studied before in this domain."

"I evaluated three LLMs — GPT-4o-mini, Mistral, and Qwen — on 9,218 query-response pairs generated from 749 real and synthetic interviews in English and Dutch. I classified hallucinations using three taxonomies, tested four automated detection methods, and discovered that GPT-4o-mini's high hallucination rate (20.77%) is largely explained by safety guardrails refusing to interpret personal data — a novel finding that has implications for how we measure hallucination in privacy-sensitive AI systems."

Q: Why GPT-4o-mini, Mistral, and Qwen — why not GPT-4 full or Llama?

- These three represent three distinct tiers: a commercial small model (GPT-4o-mini), a capable open-weight model (Mistral-Small-24B), and a Chinese open-weight model (Qwen2.5-7B)
- This gives cost/capability diversity: GPT-4o-mini is cheap to run at scale; Mistral and Qwen are self-hostable
- GPT-4o (full) would be interesting as an upper bound — I plan to add it if time permits
- Llama-3.1-8B is a natural next step for open-source reproducibility
- The current 3-model selection is standard for IEEE papers in this space — sufficient to show cross-model patterns

Q: Is 749 interviews a sufficient sample size?

- 749 interviews × 6 query types × 3 models = 9,218 response pairs — this is the evaluation corpus, not a training set
- For detection comparison (RQ2), no training is needed — detectors are pre-trained models evaluated on our data
- The 95% Wilson CIs are narrow: GPT-4o-mini rate is 20.77% ± 1.4% — statistically robust
- McNemar's test on 3,000+ shared samples gives very high statistical power
- IEEE papers in hallucination detection typically use 500–5,000 evaluation instances — we are in that range

Q: Your synthetic data is AI-generated — how do you validate its quality?

- The synthetic interviews were generated to match the structure of real Convo interviews — same JSON schema, same utterance format
- The hallucination rates are nearly identical across real and synthetic interviews — suggesting the models respond similarly to both
- The EN/NL language distribution in synthetic data matches the real data proportions
- This is a known limitation — I address it explicitly in the Threats to Validity section as a future improvement
- A human quality check on 50 synthetic interviews would strengthen this — I can do that alongside annotation

Q: GPT-4o-mini is evaluating outputs that partly came from GPT-4o-mini — isn't that circular?

- This is a known limitation called 'LLM-as-judge bias' — I address it explicitly in the paper
- The mitigation is manual annotation + Cohen's Kappa: 200 human-labelled samples will validate whether the auto-labels are trustworthy
- Prior work (Zheng et al., 2023; Manakul et al., 2023) uses the same LLM-as-judge approach — it is standard practice
- GPT-4o-mini was the evaluator for ALL three models including Mistral and Qwen — so the circular concern only applies to ~33% of responses
- Using a separate evaluator (e.g. Claude) for cross-validation is in the future work section

Q: Why the RAGTruth taxonomy with 7 types — why not just HALLUCINATED/FAITHFUL?

- A binary label tells you THAT a model hallucinates, not WHY or HOW
- The 7 types reveal important patterns: 60% of GPT-4o-mini errors are REFUSAL — this would be invisible with binary labels
- Different error types have different remediation strategies: TEMPORAL_CONFUSION needs better chunking; REFUSAL needs a different prompt
- Multi-class taxonomy is standard in ACL/EMNLP hallucination papers — binary is considered too coarse for research contributions
- I compare all three taxonomies on the same data — the comparison itself is a contribution

Q: SelfCheckGPT returned F1=0 — does that mean your detection failed?

- SelfCheckGPT didn't fail — it failed for a specific and interesting reason that becomes a finding
- For interview data, all 5 sampled responses tend to be consistently wrong in the same way (all refuse to interpret sentiment)
- BERTScore sees high consistency between samples, so it scores hallucination risk as low — a false negative
- This shows SelfCheckGPT's limitation in dialogue domains: it assumes inconsistency = hallucination, but models can be consistently wrong
- I report this as a finding: SelfCheckGPT is not suitable for interview RAG due to correlated refusal behaviour

Q: What is specifically novel about this versus existing RAG hallucination papers?

- First study on interview/dialogue domain RAG — all prior work uses Wikipedia, news, or general QA datasets
- Refusal-hallucination distinction: we identify safety guardrails as a separate failure mode — not studied before
- Multilingual: English + Dutch — almost all RAG hallucination papers are English-only
- Multi-taxonomy comparison on identical data: we compare RAGTruth vs Huang vs DiaHaLu on the same responses
- Predictive hallucination detection: hallucination_predictor.py predicts risk before generation from retrieval signals

Q: Why interview data specifically — is this just a domain application paper?

- Interview data has unique properties that make it more than a simple application: privacy sensitivity, personal content, emotional nuance
- The safety guardrail problem (REFUSAL_HALLUCINATION at 60%) only emerges in privacy-sensitive domains — you would not see this in Wikipedia RAG
- Interview data is increasingly used in qualitative research and market research — the practical impact is real
- The multilingual aspect (EN + NL) adds a research dimension beyond domain application
- The three taxonomy comparison adds methodological contribution beyond the domain

Q: What's the practical impact — who would actually use this?

- Researchers and companies using AI-powered interview tools (like Convo) need to know when to trust AI-generated summaries
- Market research firms running hundreds of customer interviews could use hallucination detection to flag unreliable summaries automatically
- HR platforms using AI to analyse candidate interviews face the same problem — misquoting a candidate has legal/ethical consequences
- The correction_loop.py module demonstrates a practical fix: detect hallucination, auto-correct it
- The hallucination_predictor.py module enables pre-emptive quality filtering before showing results to users

Q: What would you do with more time?

- Add GPT-4o (full) as upper-bound model and Llama-3.1-8B for open-source reproducibility
- Complete AlignScore and RAGAS experiments for all models, then train and evaluate the ensemble detector
- Expand manual validation from 200 to 500+ samples with a second annotator for robust Kappa
- Run the hyperparameter sweep to find optimal chunk_size + top_k configurations
- Study the correction_loop performance — how often does auto-correction actually fix hallucinations?

Section 10 — What More to Explore

10.1 High-Priority Gaps to Fill Before Submission

#	Gap	How to Fill It	Time Needed
1	AlignScore + RAGAS missing for all 3 models	python3 src/run_all.py --steps rq2 (these should run if transformers installed)	2–4 hours
2	Mistral has no RQ2 results	python3 src/run_all.py --models mistral --steps rq2	1–2 hours
3	Manual annotation (0/198 rows labelled)	Follow Section 7 of this guide. 7 sessions of 30 rows each.	~4 hours total
4	Cohen's Kappa not computed	Follows automatically from annotating. Run calculate_kappa.py after.	5 minutes
5	Ensemble detector not trained	python3 RAG_THESIS/advanced/ensemble_detector.py (needs manual labels first)	10 minutes

10.2 Additional Models Worth Testing

Model	Why Add It	Priority	How to Add
GPT-4o (full)	Upper bound — shows max possible quality and compares to GPT-4o-mini	High	Add 'gpt-4o' to DEFAULT_MODELS in run_all.py
Llama-3.1-8B	Open-source — other researchers can reproduce without API keys	High	Add 'meta-llama/Llama-3.1-8B-Instruct' and configure Ollama or HuggingFace
mBERT / XLM-R	Multilingual detection model for cross-lingual hallucination scoring	Medium	Use as AlignScore alternative in experiment_rq2.py
Gemini 1.5 Flash	Google alternative — multilingual, cost-efficient	Low	Requires Gemini API key

10.3 Detection Method Improvements

Current Method	Improvement	Expected Impact	Effort
AlignScore (RoBERTa-base)	Upgrade to RoBERTa-large or DeBERTa-v3-large	+5–8% F1 based on literature	Low (change model name in code)
SelfCheckGPT (BERTScore)	Try MQAG variant (multiple-choice QA)	Better precision for factual claims	Medium (add MQAG dependency)
RAGAS (faithfulness only)	Add RAGAS answer_relevancy metric	Broader hallucination coverage	Low (already in ragas library)
Fixed ensemble weights	Use logistic regression weights from ensemble_detector.py	Learned weights beat fixed 30/40/30	Low (run ensemble_detector.py)

Current Method	Improvement	Expected Impact	Effort
5th ensemble signal	Add depth_scoring.py as 5th feature in ensemble	Unique to your work	Low (depth_scoring.py exists in src/)

10.4 Advanced Modules Not Yet Run

Module	What It Does	How to Run	What to Report
hallucination_predictor.py	Predicts hallucination risk BEFORE the response is generated, using retrieval-time signals: similarity score, chunk size, number of retrieved chunks, query type	python3 RAG_THESIS/advanced/hallucination_predictor.py	AUC-ROC of predictor vs actual hallucination rate. If AUC > 0.7, the model can be used to pre-filter risky queries.
correction_loop.py	Detects high-hallucination responses and re-queries the model with stronger grounding instructions to produce a corrected response	python3 RAG_THESIS/advanced/correction_loop.py	% of hallucinations corrected. MiniCheck score before vs after correction.
retrieval_correlation.py	Computes Pearson/Spearman correlation between retrieval quality metrics (similarity score, chunk size) and hallucination rate	python3 RAG_THESIS/advanced/retrieval_correlation.py	Correlation coefficient. If high, retrieval quality predicts hallucination — useful finding.
run_no_rag_baseline.py	Answers all queries without providing the transcript as context (no RAG). Shows what happens if you just ask the model directly.	python3 RAG_THESIS/ablation/run_no_rag_baseline.py	Hallucination rate without RAG vs with RAG. Should be much higher without RAG — proves RAG helps.

10.5 Future Research Directions

Human-in-the-loop correction: Build a UI where researchers see hallucination alerts and approve/reject corrections before viewing interview summaries

Fine-tuning a domain-specific detector: Use the 9,218 labelled responses to fine-tune a compact NLI model specifically for interview hallucination — faster and cheaper than MiniCheck

Real-time detection: Integrate hallucination detection into the live interview pipeline so researchers are warned before data is saved

Cross-dataset transfer: Test whether detectors trained on Convo interviews generalise to other interview platforms (academic research, HR, medical)

Dutch-specific evaluation: Partner with Dutch researchers to add more NL interviews and a Dutch-speaking annotator for cross-lingual Kappa